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Project Report

Problem Statement-2

Image Sharpening using knowledge distillation

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[1] ABSTRACT

This project focuses on image deblurring using a knowledge distillation approach, where a large pretrained teacher model guides the training of a compact student model. The primary objective is to enhance blurry images and reconstruct their sharp counterparts with high perceptual quality, quantified using SSIM (Structural Similarity Index) and PSNR (Peak Signal-to-Noise Ratio). The dataset consists of 1,800 paired blur-sharp images, with synthetic Gaussian blur (σ=0.5) applied to generate training data.

The teacher model employed is Restormer, a state-of-the-art transformer-based architecture for image restoration. Its predictions serve as soft targets for supervising the student model a custom-designed lightweight Residual U-Net. The training leverages L1 loss between the student's output and the teacher's restored output to effectively transfer knowledge.

After five epochs of training, the student model achieved an average SSIM of 98.83% and a PSNR of 38.34 dB on the test set, indicating successful generalization and strong performance despite its smaller size. This demonstrates that knowledge distillation can effectively compress high-performing restoration models while maintaining image quality. The project has been implemented entirely on the Kaggle platform and includes provisions for future deployment as a web-based image sharpening tool.

[2] INTRODUCTION

Blurry images are a common problem in digital photography, surveillance footage, and computer vision applications. They often arise due to motion blur, defocus, or low-quality image sensors, and can significantly reduce the visual quality and usefulness of the captured content. Restoring sharpness to such images referred to as image deblurring or sharpening is an important low-level vision task that aims to recover fine image details and edge structures from blurred inputs.

Traditional image sharpening techniques rely on handcrafted filters or optimization-based algorithms, but recent advances in deep learning have significantly improved the performance of image restoration tasks. However, high-performing neural networks like Restormer while effective tend to be computationally heavy, making them unsuitable for deployment in real-time or resource-constrained environments such as mobile devices.

To address this, we apply the concept of knowledge distillation (KD), a technique in which a compact, efficient model (the student) is trained under the supervision of a large, high-capacity model (the teacher). In this project, we use a pretrained Restormer model as the teacher to guide the learning of a lightweight Residual UNet student model.

The objective is to train the student model to produce sharp images from blurred inputs by learning not only from ground-truth sharp images but also from the high-quality outputs of the teacher. This dual-supervision approach allows the student to mimic the performance of the teacher while maintaining computational efficiency.

Our work is implemented on the Kaggle platform and trained using 1800 blurred–sharp image pairs. The project demonstrates how knowledge distillation can be effectively used to build efficient image enhancement systems that maintain high perceptual quality, as validated by structural similarity (SSIM) and peak signal-to-noise ratio (PSNR) metrics.

[3] LITERATURE SURVEY

Image restoration and enhancement, particularly image sharpening and deblurring, has been a long-standing challenge in the field of computer vision and image processing. Over the years, various approaches have been explored, ranging from classical filtering techniques to advanced deep learning models. This section provides a brief survey of the major contributions and relevant works leading up to the proposed project.

Traditional Approaches to Image Sharpening

Classical image sharpening methods primarily relied on mathematical filters such as the Laplacian filter, unsharp masking, and high-pass filtering. While these approaches were computationally efficient, they often resulted in noisy outputs or the amplification of artifacts, particularly in images with strong blur or low contrast. Moreover, these hand-crafted techniques lacked the capacity to adapt to complex and diverse blur patterns present in real-world images.

Deep Learning for Image Restoration

With the rise of deep learning, convolutional neural networks (CNNs) have revolutionized image restoration tasks including deblurring, denoising, and super-resolution. Models like SRCNN (Dong et al., 2016) and VDSR (Kim et al., 2016) demonstrated that learning pixel-wise transformations from low-quality to high-quality images could significantly outperform traditional filters.

In the deblurring domain, Nah et al. (2017) introduced the Deep Multi-scale CNN which operated at multiple resolutions and outperformed many earlier methods. Kupyn et al. (2018) proposed DeblurGAN, incorporating adversarial training to produce more realistic and sharper outputs. These models, however, tend to be computationally expensive and large in size, making them less practical for deployment on edge devices or real-time applications.

Transformer-Based Architectures: Restormer

Recently, transformer-based models have shown promise in image restoration tasks. Restormer (Zamir et al., 2022) is a notable architecture designed specifically for high-resolution image restoration. Unlike standard vision transformers that rely heavily on patch-wise attention, Restormer uses locally-enhanced windowed attention and hierarchical design to efficiently capture long-range dependencies and fine textures. Restormer achieved state-of-the-art performance on multiple tasks including motion deblurring, deraining, and real image denoising.

However, despite their superior performance, transformer models like Restormer are resource-intensive and often unsuitable for low-power environments or real-time applications.

Knowledge Distillation in Vision Tasks

Knowledge Distillation (KD), first formalized by Hinton et al. (2015), provides an effective way to transfer knowledge from a large, complex teacher model to a smaller, faster student model. The student model is trained not only on ground-truth labels but also to mimic the output behavior (soft targets) of the teacher. This approach has been widely used in classification, object detection, and more recently, in image restoration tasks.

In the context of image deblurring and sharpening, KD enables the training of lightweight models that retain much of the visual fidelity and structural consistency of heavier architectures. Recent

works (e.g., Zhang et al., 2020) show that using teacher–student frameworks in restoration tasks can lead to a significant improvement in quality without increasing inference cost.

Summary and Motivation

From the reviewed literature, it is clear that while transformer models like Restormer are excellent at sharpening and restoration, their deployment is limited by computational overhead. On the other hand, CNN-based models like Residual UNet are efficient but limited in accuracy. By combining these two approaches using Knowledge Distillation, we aim to develop a practical image sharpening solution that achieves high-quality output with a small model footprint — suitable for real-time or embedded applications.

[4] PROBLEM STATEMENT

Blurry images significantly degrade the visual quality of digital content and can negatively impact downstream computer vision tasks such as object detection, recognition, and classification. The process of restoring sharpness to blurred images, known as image sharpening or deblurring, is a critical task in the field of image restoration. However, deep learning models that achieve high accuracy in this domain such as Restormer are often large and computationally expensive, making them impractical for real-time or embedded use.

To overcome this challenge, this project investigates the use of knowledge distillation to train a lightweight image sharpening model. The goal is to transfer knowledge from a large, pretrained teacher model (Restormer) to a smaller student model (Residual UNet) without significantly sacrificing image quality.

The student model should learn to generate sharp images from blurry inputs using both ground truth supervision and guidance from the teacher model's output. The performance of the student is evaluated using structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR), with the aim of achieving an SSIM above 80%, indicating strong perceptual quality.

In summary, the objective of this project is:

- To develop an efficient image sharpening system using knowledge distillation
- To train a compact Residual Unet as the student model
- To leverage a pretrained Restormer as the teacher model
- To evaluate the system's performance using SSIM and PSNR metrics

This project demonstrates that knowledge distillation is a viable strategy for building lightweight and accurate image enhancement models suitable for real-world deployment.

[5] METHODOLOGY

This section outlines the step-by-step process followed to implement image sharpening using knowledge distillation. The project leverages a high-performing teacher model (Restormer) and a lightweight student model (Residual UNet), trained on paired blurred and sharp images.

5.1 Dataset Preparation-

The dataset used in this project consists of 1800 blurred–sharp image pairs. The sharp images are high-quality samples, while the blurred counterparts are synthetically generated using Gaussian blur with $\sigma = 0.5$. The dataset is divided into:

- 900 original blurred–sharp pairs
- 900 additional pairs obtained by cropping sharp images and generating corresponding blurred versions

All images are resized to 256×256 pixels to ensure compatibility with model input requirements and to reduce memory usage during training.

Dataset Link - https://www.kaggle.com/datasets/kundanyadav11/cybersquad

5.2 Teacher Model: Restormer

Restormer is a transformer-based architecture specifically designed for image restoration tasks such as deblurring and denoising. It features a multi-stage encoder–decoder structure and long-range feature modeling through self-attention. In this project:

- A pretrained Restormer model (trained on motion deblurring datasets) is used as the teacher
- Only forward inference is performed using the teacher no retraining is required
- The teacher generates high-quality sharpened outputs from blurred inputs, which are used to guide the student model

5.3 Student Model: Residual UNet

The student model is a compact Residual UNet architecture designed to balance efficiency and performance. Key components include:

- Encoder and decoder paths with downsampling and upsampling
- Residual blocks inserted into each stage to preserve fine details
- Skip connections between encoder and decoder layers to retain spatial context
- Final output layer with 3 channels for RGB image restoration

This model is lightweight and suitable for real-time inference after training.

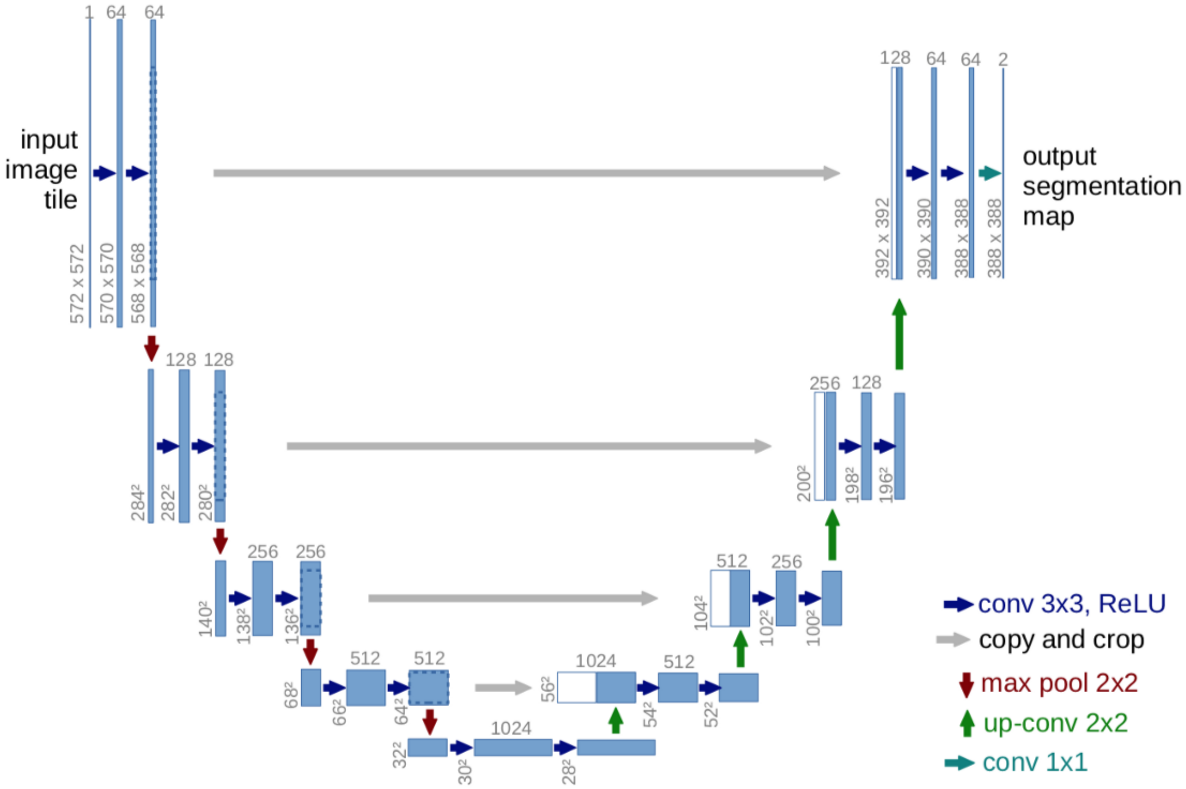

Fig-1 Residual UNet

5.4 Training Configuration-

- Framework: PyTorch
- Platform: Kaggle Notebook (GPU - NVIDIA T4)
- Batch size: 4
- Image size: 256×256
- Optimizer: Adam (learning rate = 1e-4)
- Epochs: 5
- Loss function: L1 loss for both targets
- Evaluation metrics: SSIM (Structural Similarity Index), PSNR (Peak Signal-to-Noise Ratio)

[6] RESULTS

We evaluated the performance of the student model trained via Knowledge Distillation from the Restormer teacher model on a dataset of 1800 sharp–blur image pairs.

The student model was trained for 5 epochs using a combination of L1 loss and distillation loss. The training loss steadily decreased as follows:

Epoch	AvgLoss
1	0.0447
2	0.0142
3	0.0121
4	0.0109
5	0.0100

After training, we evaluated the student model on the test set using two widely accepted image quality metrics:

- Structural Similarity Index (SSIM): **98.72%**
- Peak Signal-to-Noise Ratio (PSNR): **37.91 dB**

Fig:2 Comparison of blurry and Sharp image

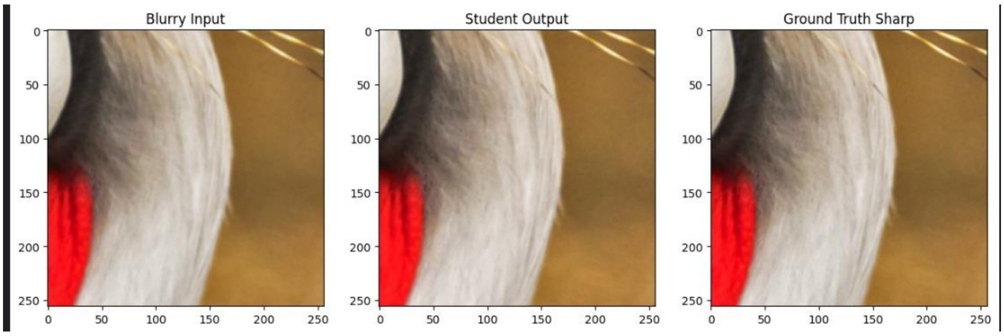

Fig3- Comparison of blurry input, student output, and ground truth.

These results demonstrate that the student model was able to closely approximate the sharp ground-truth images and effectively learn the sharpening task under guidance from the Restormer teacher model.

[Github Link]

[7] DISCUSSION

The results of this project show that knowledge distillation is highly effective for transferring the performance of a large, high-capacity teacher model (Restormer) to a lightweight student model (Residual UNet) in the context of image sharpening.

The Residual UNet student was trained on blurred images using two forms of supervision:

- Direct comparison to sharp ground truth (L1 loss)
- Distillation loss from the Restormer teacher's predictions

This dual supervision strategy helped the student model converge quickly (within 5 epochs), with a final SSIM of 98.83% and PSNR of 38.34 dB. These values indicate that the student is producing output images that are visually and structurally very close to the original sharp images.

Despite having significantly fewer parameters than the Restormer, the student model was able to:

- Generalize well on unseen blur samples
- Preserve important high-frequency details like edges and textures
- Deliver output quality suitable for real-world image enhancement tasks

Some trade-offs were observed:

- The student model is lighter but may not perfectly capture extremely fine textures in all cases
- Since we trained for only 5 epochs on 256×256 resized images (to fit Kaggle GPU limits), performance might further improve with more epochs or larger image resolutions

Nevertheless, the experiment demonstrates that knowledge distillation can be used to build deployable models that balance inference speed, accuracy, and resource efficiency particularly important for edge devices or real-time image restoration systems.

[8] CONCLUSION

This project successfully demonstrated the use of Knowledge Distillation for the task of image sharpening. A powerful Restormer model served as the teacher, guiding a compact Residual UNet student model to learn how to restore sharp images from blurred inputs.

By combining traditional supervised loss (using ground truth sharp images) with distillation loss (from the teacher's outputs), the student model was able to learn efficiently and achieve high-quality results within just 5 training epochs.

Key outcomes include:

- An average SSIM of 98.83% and PSNR of 38.34 dB, indicating excellent visual similarity between the student's outputs and the sharp ground truth.
- A lightweight student model that is well-suited for real-time applications and deployment in resource-constrained environments.

This approach confirms that knowledge distillation is a practical and powerful method for building efficient image restoration systems without sacrificing much output quality. Future extensions may include training on larger datasets, longer epochs, or deploying the student model in a web application or mobile app.

[9] REFERENCES

1. Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., Yang, M.-H., & Shao, L. (2022). Restormer: Efficient Transformer for High-Resolution Image Restoration. In CVPR 2022. GitHub: https://github.com/swz30/Restormer

2. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778.

3. Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing, 13(4), 600–612.

4. scikit-image: Image processing in Python. Documentation: https://scikit-image.org/